SERANGOON JUNIOR COLLEGE
2009 JC2 PRELIMINARY EXAMINATIONS

MATHEMATICS
Higher 2 **9740/Paper 2**

Monday **24 Aug 2009**

Additional materials: Writing paper
List of Formulae (MF15)

TIME : 3 hours

READ THESE INSTRUCTIONS FIRST
Write your name and class on the cover page and on all the work you hand in.
Write in dark or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

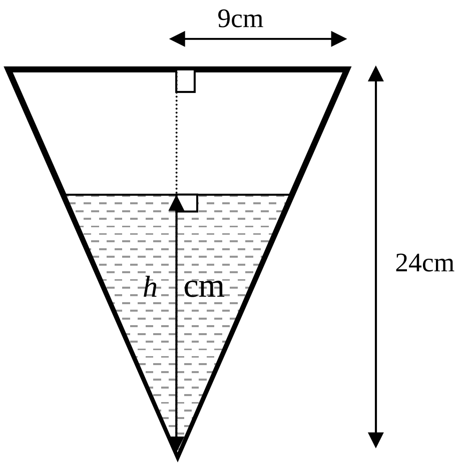
Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets **[ ]** at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This question paper consists of 7 printed pages.

Section A: Pure Mathematics [40 marks]

1.

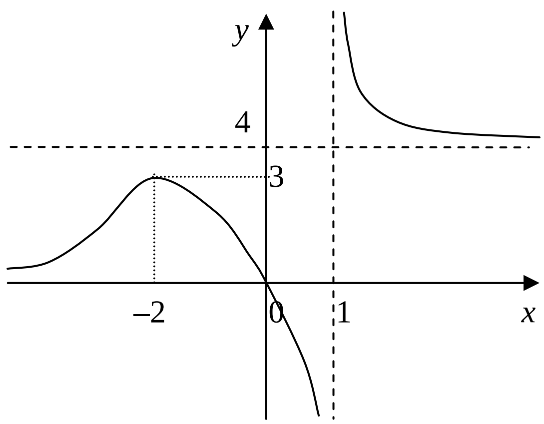


The diagram shows a vertical cross-section of a container in the form of an inverted cone of height 24 cm and base radius 9 cm. The container is initially empty. Water is poured into the container at a constant rate of $30 \text{ cm}^3\text{s}^{-1}$ and the height of water level in the container at time t is h cm.

(i) Show that the volume of water in the container is given by $V = \frac{3}{64}\pi h^3$. [2]

(ii) Find the value of h at the instant when the rate of increase in the depth of water is 5.65 cms^{-1}, leaving your answer in 2 decimal places. [3]

2(a)



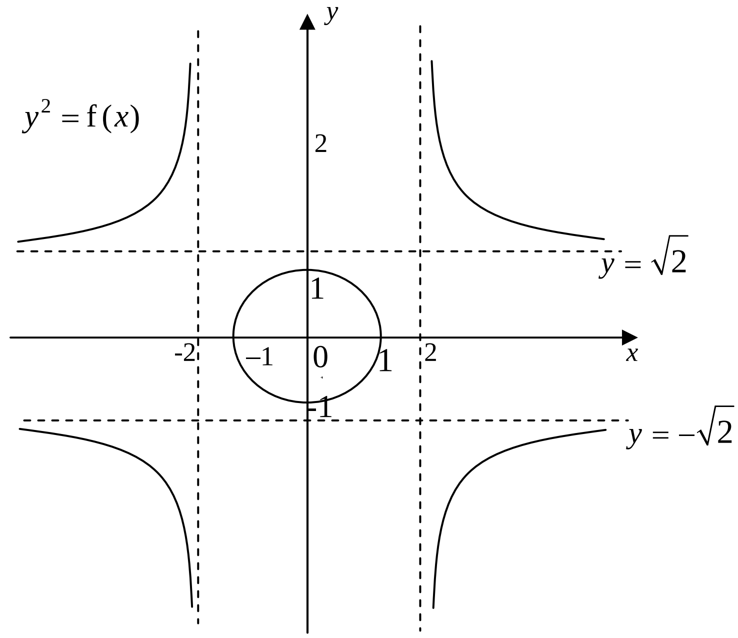
The graph of $y = \mathrm{f}(x)$ is shown above. Sketch on separate diagrams, the graphs of

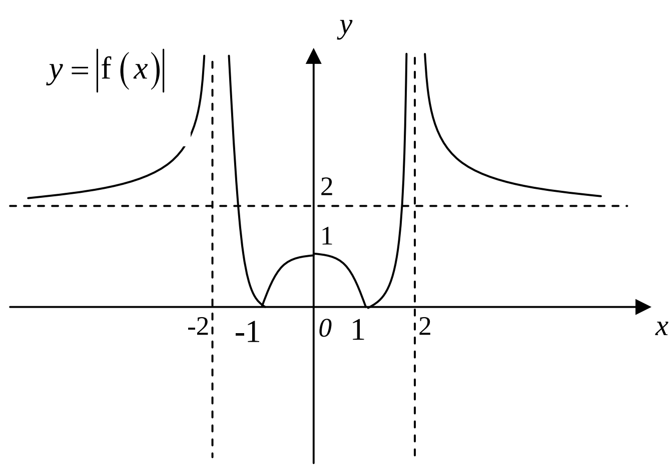
(i) $y = \dfrac{1}{\mathrm{f}(x)}$ [2]

(ii) $y = \mathrm{f}'(x)$ [2]

You are expected to indicate clearly all important features of the graphs in each case.

(b) The graphs of $y^2 = \mathrm{f}(x)$ and $y = |\mathrm{f}(x)|$ are shown below.





Sketch the graph of $y = \mathrm{f}(x)$, indicating clearly all important features of the graph. [2]

3. A curve C is defined parametrically by the equations $x = t - \frac{1}{t}$, $y = t + \frac{1}{t}$, $t \neq 0$.

(i) Show that the equation of normal at a point with parameter t is given by
$y = -\frac{t^2+1}{t^2-1}x + \frac{2(t^2+1)}{t}$. [3]

(ii) Hence, determine the equation of normal at the point $t = 2$. [1]

(iii) The normal at the point $t = 2$ cuts the curve C again at the point Q. Determine the coordinates of Q. [4]

4. The polynomial $\mathrm{P}(z)$ has real coefficients. The equation $\mathrm{P}(z) = 0$ has a root $3e^{i\theta}$ where $0 < \theta < \pi$.

(i) Show that a quadratic factor of $\mathrm{P}(z)$ is $z^2 + az\cos\theta + b$ where a and b are real numbers to be determined. [3]

(ii) Solve the equation $z^4 + 81 = 0$, expressing the solution in the form $re^{i\theta}$, giving the values of r and θ in exact form, where $r > 0$, and $-\pi < \theta \leq \pi$. [4]

(iii) Hence or otherwise, show that $z^4 + 81$ can be expressed in the form $\left(z^2 - p\left(\sqrt{2}z\right) + q\right)\left(z^2 + p\left(\sqrt{2}z\right) + q\right)$ where p and q are real numbers to be determined. [3]

5. A plane π_1 has equation $\mathbf{r}.\begin{pmatrix} -2 \\ -1 \\ 5 \end{pmatrix} = 3$. The points A and B have coordinates $(2, -1, 3)$ and $(4, 1, -3)$ respectively.

(i) Find the exact length of projection of the vector $\overrightarrow{AB}$ onto the normal of the plane π_1. [2]

(ii) Given that plane π_1 meets another plane π_2 with equation $\mathbf{r}.\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 2$ along the line l, find a vector equation of the line l in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$. [2]

(iii) A third plane π_3 contains the line l and passes through the point with position vector $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Find the vector equation of the plane π_3 and express it in the form $ax + by + cz = d$. [4]

(iv) Deduce, or prove otherwise, that the system of equations

$$-2x - y + 5z = 3$$
$$x + y + z = 2$$
$$4x + 3y - 3z = 1$$

has an infinite number of solutions, stating your reason(s) clearly. [2]

(v) The plane π_4 is parallel to π_1 and has equation $-2x - y + 5z = 5$.

Comment on the geometrical representation of the 3 planes π_2, π_3, and π_4. [1]

Section B: Statistics [60 marks]

6. A presidential candidate in country USS wanted to compare the preference of registered voters in the north-eastern states. The candidate hired a professional pollster to randomly choose 1000 registered voters to be interviewed on their preference.

(i) Describe in detail how the method of quota sampling can be used to select the 1000 registered voters for the interview. [2]

(ii) The pollster decided to use stratified sampling. List any two pieces of information he should consider acquiring. [2]

7. A box contains 20 balls that are labeled from 1 to 20. Raju and Kelvin take turns to pick a ball from the box with Kelvin drawing a ball first. The game will stop when a prime number (2, 3, 5, 7, 11, 13, 17, 19) is drawn.

(a) If the balls are drawn **without replacement**, find the probability that Kelvin is the first to pick a prime number given that the game ends on the second draw of the player. [3]

(b) If the balls are now drawn **with replacement**, find the probability that Raju is the first to pick a prime number. [3]

8. In a factory, a machine automatically cuts out metal caps. It is given that 0.5% of the caps are dented. Find the probability that out of 200 caps, at least 4 are dented.
A box containing 200 of such metal caps are analyzed. The box is considered 'substandard' if it contains at least 4 dented caps. Using a suitable approximation, calculate the probability that at most 6 out of 300 boxes will be \`substandard'. [6]

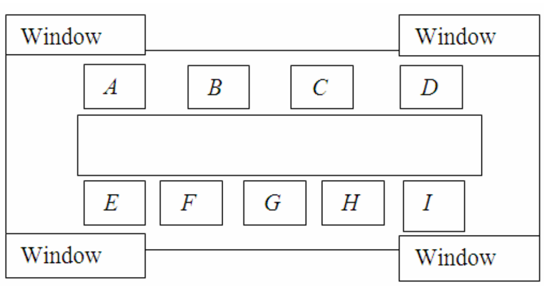
9. The random variable X has a normal distribution with mean μ and variance σ^2. The random variable S is the sum of 10 independent observations of X, and the random variable T is the sample mean of 4 independent observations of X. It is given $\mathrm{P}(S > 510) = 0.132$ and $\mathrm{P}(T < 53) = 0.983$. Find μ and σ. [8]

10. A fruit grower produces a large number of apples every day. A small proportion p of these apples is infected. A check is carried out each day by taking a random sample of 50 apples and examining them for infection. The number X of infected apples in the sample of 50 apples may be assumed to have an approximate Poisson distribution.
    (i) State an inequality satisfied by p. [1]
    (ii) The probability that none of the 50 apples is infected is 0.3. Show that the probability that not more than 2 apples are infected is 0.879. [3]
    (iii) If exactly 3 apples are infected, a further random sample of 10 apples is taken. The day's production is accepted as satisfactory in either of the following two cases :
    a) the number of infected apples in the sample of 50 is at most 2;
    b) the number of infected apples in the sample of 50 is 3 and the number of infected apples in the sample of 10 is 0 or 1.
    Find the probability that the day's production is accepted as satisfactory. [4]

11. (a) In how many ways can 7 packs of chocolates be distributed among 20 children, if no child can get more than one pack? [1]

    (b) In how many ways can 4 different gifts be distributed among 20 children if each child can get any number of gifts? [1]

    (c) 9 people go to a restaurant with the layout given by the diagram below. In how many ways can the 9 people be seated on the chairs marked *A*, *B*, *C*, …, *H*, *I*? [1]



    Find the number of ways in which the 9 people can be seated if
    (i) 3 particular people do not want to be seated on any of the 4 chairs *A*, *D*, *E* and *I*, in front of windows. [2]
    (ii) 2 particular people do not want to be seated next to each other on the same side of the table. [3]

12. A film distributor claimed that the tickets sales for a particular movie hit an average of 1185 tickets a day. A random sample of 56 days was taken and the sales, x tickets, for each day were recorded. It was found that

$$\sum(x-1000)=9976 \text{ and } \sum\left(x-\bar{x}\right)^2=43286.$$

(i) Find, correct to 2 decimal places, the unbiased estimate of the population mean and variance of the ticket sales. [2]

(ii) Test at 2% significance level whether the distributor was overstating his claim. [5]

(iii) Explain what is meant by the phrase 'at 2% significance level' in the context of this question. [1]

(iv) The same sample of 56 days is used in another test to determine whether the average sale of tickets is 1185. If the null hypothesis is rejected, determine an inequality satisfied by the significance level $\alpha\%$, of this test. [2]

13. The temperature of the body, $\theta\ {}^{\circ}C$, is observed to vary with time t. At a particular environmental temperature $\theta_0{}^{\circ}C$, the table below shows a set of data relating $Y(\theta-\theta_0)$ and time t.

$Y\ ({}^{\circ}C)$	60	46	33	21	15	11	8	7
t (min)	1	4	7	10	13	16	19	22

(i) Sketch a scatter diagram of Y against t for the data and, by comparing the product moment correlation coefficients of the models below, deduce which of the following models is more appropriate.

A: $Y=a+bt^2$
B: $\ln(Y)=a+bt$ [4]

(ii) For the selected model, find the values of a and b. [2]

(iii) If the environmental temperature is $\theta_0=28\ {}^{\circ}C$, find the temperature of the body, correct to 1 decimal place, half an hour after the start of the experiment. Comment on why it is unwise to use the selected equation in (i) to estimate this temperature of the body. [2]

(iv) Explain why it is reasonable to use the selected regression line in (i) to estimate t when $Y=50$. [2]

[Turn Over